

torch-spyre Ecosystem Health Report

May 20, 2026 | Pod: rganti-spyre-main | PyTorch 2.11.0

3/3 BUILD PASS	3/4 CI PASSING	20 PRS MERGED (7D)	30 OPEN BUGS
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Build Status			
Component	Commit	Status	Notes
deeptools	7325fd8d0e — int32→int64 cast conversions	PASS	Incremental build
flex	8263267 — Upgrade to C++20	PASS	Full install
torch-spyre	cd0d529 — Per-test per-op tags utility	PASS	Editable wheel, ~1m 17s

CI Status			
Workflow	Last Run	Status	Trend
Spyre Hardware Tests	May 20	PASS	Healthy
Linters	May 20	PASS	Healthy
Upstream PyTorch Tests	May 20	PASS	Healthy
Nightly Tests	May 20	FAIL	Persistently failing (3+ days)

Nightly Failure Triage	
Failure 1: test_instantiate_device_type_tests_mro Error: AssertionError — expected _TestMROCheckPRIVATEUSE1, got _TestMROCheckSPYRE Root cause: PyTorch 2.11 uses registered device name instead of generic PRIVATEUSE1 Duration: Since May 16 Assessment: Regression (PT upgrade) Fix: Branch fix-mro-test-pt211 (commit 81dcb4e) — needs merge to main	
Failure 2: test_activation_fn_mish_fp16_lx_planning_reduction	

Error: Compiled vs CPU mismatch — 1/128 elements off by 0.18, tolerance 0.01
Root cause: PR #2071 changed seed strategy; new inputs hit fp16 precision boundary
Duration: Since May 19 | **Assessment:** Deterministic precision edge case
Fix needed: Widen atol for mish+reduction combo or adjust seed

Version Alignment

Check	Expected	Actual	Status
PyTorch	~=2.11.0 (pyproject.toml)	2.11.0+cpu	MATCH

Repo Drift (Last 7 Days)

Repository	Total Commits	API-Touching	Risk
deeptools	90	10	HIGH
flex	17	9	HIGH
torch-spyre	59	—	—

Key upstream changes: flex C++20 upgrade, warnings-as-errors enabled; deeptools EmitBinaryOptions refactor, new dfmt_send/rec signatures.

Blocking PRs

PR	Title	Why Blocking	Review	Age
#1764	Add SpyreCode to JobPlan translation	Cross-repo interface (flex/deeptools)	REVIEW REQUIRED	27 days

Additionally: 9 dependabot PRs open (actions/checkout v6, actions/cache v5, setup-uv v8, etc.)

Project Velocity

20 merged (7d) 20 in review 19 stalled (>7d)

Velocity is strong at ~3 merges/day. Notable stalled: #816 Multi-Spyre Device (82 days), #1721 approved but unmerged (29 days).

Open Bugs

Metric	Value	Trend
Total open	30	Growing — 0 closures in 12 days
New (24h)	1	#2166 softmax sub-optimal performance
Filed (7d)	+3	#2166, #2087, #2070

Closed (7d)0Fix velocity stalled

Smoke Test

Test	Result
Import & device detection	PASS
Eager execution (fp16 add)	PASS
Compiled execution (inductor)	PASS

Action Items

1

Merge branch **fix-mro-test-pt211 to main** — fixes nightly MRO test failure (blocking 4+ days)

2

Fix mish **fp16 tolerance** in `test_inductor_ops_lx_planning.py` — widen atol or adjust seed (failing since May 19)

3

Address bug **fix velocity** — 30 open bugs, 0 closures in 12 days; triage and assign top items

4

Merge **PR #1721** — approved 29 days ago, just needs final merge action

5

Batch merge **dependabot PRs** — 9 open, risk of GitHub deprecating older action versions

6

Monitor **upstream risk** — 90 deeptools + 17 flex commits with header/API changes this week